

CHAPTER 3.9

Textiles

Marie-Louise Nosch

Introduction

Textiles play a major role in the Mycenaean palace economies and continue to be indispensable to the ancient economy of the Early Iron Age (EIA). Recent research has focused on Aegean Late Bronze Age (LBA) textile technology based on textile tools, archaeological textiles, Linear A and Linear B evidence for textile types and the organization of textile production (Barber 1991; Nosch and Laffineur 2012). The introduction of wool in the Early Bronze Age (EBA) resulted in remarkably colorful clothing, achieved through the systematic cultivation of plant dyes and exploitation of murex dyes. New evidence for textiles, clothing, and related technology and terminology from Bronze Age (BA) Near Eastern, Levantine and Egyptian archaeology, epigraphy and iconography is currently emerging (Michel and Nosch 2010; Breniquet and Michel 2014; Harlow, Michel, and Nosch 2014). However, the textiles, clothing cultures and textile tools dating from the LBA/EIA transition and the EIA are, regrettably, less known (Marinatos 1967; Wagner-Hasel 2000; Burke 2010). The field of textiles is particularly promising for the integrated study of LBA and EIA because textile technology represents a continuum throughout the period and also because through textile techniques, terminology, and tools, we can perceive changes, foreign influences, and innovations that also impact on other crafts, such as pottery decoration and the use of metal fibulae. Finally, textiles used as sails were introduced in the Middle Bronze Age (MBA) and were vital for transportation throughout the period but probably gained greater importance with trade and colonization.

Primary Evidence for Aegean Textiles

Textiles in the Aegean were made of flax, and, since the BA, also of sheep and goat wool (Andersson Strand and Nosch 2015). It is generally assumed that textiles do not survive well in the Aegean, but recent surveys and publications show both new textile finds and advanced analyses of old finds (Spantidaki and Moulhéat 2012; Kolonas et al. 2017). A wide range of techniques is attested, for example, in Akrotiri where wool threads were recovered, a fine linen tabby textile with 20–22 threads per cm², fringes, a hem with stitching, and embroidery, and another fabric that is weft-faced with a sophisticated combination of spin-pattern made of thin S-plyed warp yarns and thicker z-spun wefts. These discoveries illustrate that, by the beginning of the LBA, a wide array of textile techniques was already in use in the Aegean. This is also evidenced by the discovery of a braided ribbon of linen, goat hair, and perhaps nettle in Late Minoan (LM) III Chania. Other LBA finds include fine tabbies of S-plyed linen yarns preserved on the surface of metal objects in graves in Mycenae, Pylona (Rhodes), Salamis, and Mochlos. However, from the turn of the millennium, we witness significant changes in both spinning and weaving traditions and technologies (Spantidaki and Moulhéat 2012: p. 194): the appearance of twills, the growing use of wool, the increase of single z-spun yarns, and the predominance of weft-faced fabrics, often made with thicker, fewer warps and many, thinner wefts. This technological complexity is evident in EIA Lefkandi where the textiles are primarily weft-faced wool tabbies of z-spun yarns, with thread counts of 10–60 threads/cm². Textiles in large quantities and of great diversity are part of burial rituals in 9th-century Athens, 8th-century Eretria, and 7th-century Corfu (Kourou 2011; Spantidaki and Moulhéat 2012). Recently, a 10th-century burial site at Stamna in Aitolia yielded bronze cauldrons preserving many different fabrics, which date to the last centuries of the 2nd millennium, thus testifying to the textile traditions of the LBA and the earliest evidence of fabrics dyed with murex in the Aegean (Figure 3.9.1) (Kolonas et al. 2017).

Acquisition of Textile Fibers and Dyes

In LBA Mycenaean Greece, textile fibers derive from the cultivation of flax (*ri-no/linon*, *Linum usitatissimum*) and the systematic breeding of sheep flocks divided by gender and age. The palace scribes at Knossos hold accounts of some 100 000 sheep in Crete, primarily for wool. To increase



Figure 3.9.1 Detail of the cover of tripodic cauldron T-587, Stamna. Source: Courtesy: Lazaros Kolonas. Copyright: Christina Margariti, Kalliope Sarri, Irene Skals, and Marie-Louise Nosch. Reproduced with permission of Christina Margariti, Kalliope Sarri, Irene Skals, and Marie-Louise Nosch. (For color version of the figure see color plate section)

fiber production, male animals are castrated and kept in separate wool producing flocks. The archaeological bone material demonstrates that this focus on wool began in the EBA, when sheep and goats were slaughtered at a later age than in the Neolithic period. This suggests that by the EBA, sheep were a source of wool and milk rather than meat. During the BA and EIA, shepherds with their specialized knowledge (the Linear B *po-me* is the later Greek *poimen* = shepherd) controlled wool quality and degrees of pigmentation.

Other LBA textile crops include dye plants, such as madder (*po-ni-ki-jo*, *Rubia tinctoria*), safflower (*ka-na-ko*, *Carthamus tinctoria*, a yellow dye) and saffron (the logogram CROCus, *Crocus sativus*), which continue in use in the EIA and as evidenced in the Homeric adjective *krokópeplos*, saffron-colored *peplos*. Plant dyes were cultivated on Mycenaean palace land together with spices (Melena 1976; Killen 1983; Nosch 2017). Murex, too, was used for textile dyes in Mycenaean palaces as recorded in two Knossian Linear B tablets, L 758 and X 976 (Nosch 2004, 2015). The use of murex dye is well attested in Middle Minoan (MM) Crete (Apostolakou, Brogan, and Betancourt 2012), and appears to have increased in the EIA, contributing to the wealth of the city of Tyre and giving the name “Phoenician” (from *phoenix*, one among several terms the Greeks used for purple) to merchants from the Levant.

The Organization of Textile Production

The Linear B records inform us about the modalities of the Mycenaean palace-regulated textile production and its administration. Each palace had complex administrative procedures to control and monitor textile production, either via taxation or *corvée* work in the villages. This involves mobilizing and controlling an extensive labor force of women and children for textile production, and men for fiber procurement (shepherds and flax cultivators), and as fullers and weavers. This method of centrally controlled mobilization, organization and monitoring of the textile production apparently stems from a Near Eastern BA tradition that the Mycenaeans shared (Michel and Nosch 2010; Breniquet and Michel 2014). This tradition continued in the 1st millennium Neo-Assyrian and Neo-Babylonian palaces, and at Gordion, where the centrally organized textile production is reflected in the large quantities of standard loom weights and in the architecture of similar workshops associated with the palace, where textile production occurred (Burke 2007, 2010). In EIA Greece, no similar organized textile production is known. It is also uncertain when the Levantine coastal cities became renowned for highly crafted textiles, such as fabrics made by Sidonian women praised in Homer. *Iliad* 6.288–292 narrates that the most precious and richly decorated fabrics possessed by the Trojan queen Hecuba were made by Sidonian textile craftswomen who had been brought from Sidon to Troy. It seems plausible that the fall of the Mycenaean palaces and textile industries enabled a shift toward new places of production and new specializations.

Textile Technology

The Aegean belongs to a north Mediterranean weaving tradition of using a low whorl drop spindle and the warp-weighted loom with loom weights. In contrast, the horizontal loom is used in the Near East and Egypt. However, these textile tools are not consistently present in the Aegean, e.g. loom weights are abundant and spindle whorls scarce in LBA Crete and Cyprus, while Mainland Greece in the same period has numerous spindle whorls and fewer loom weights. This suggests that the warp-weighted loom and the drop spindle were sometimes used in parallel with, or were sometimes supplanted by, other types of textile tools made of perishable materials (Smith and Tzachili 2012; Andersson Strand and Nosch 2015). In terms of fabrics and weave techniques, the BA Aegean conforms to the balanced or weft-faced tabby traditions also found in the Near East. This is in contrast to the

central European and Italian early use of twills and tablet weaving, which gained increasing momentum in the EIA (Gleba 2017). Tapestry probably also existed in the BA, although no specimens have come to light in the Aegean (in contrast to Tutankhamun's grave in Egypt, with its tapestry-woven lotus flowers, birds, and hieroglyphs). BA iconography of large figurative panels, as well as terminology, suggests the existence of tapestry in the Near East and Levant (Smith 2012). Such figurative tapestries also appear in descriptions of Homeric elite households (Wagner-Hasel 2000).

The primary BA loom weight types are spherical, pyramidal or discoid; they are mostly well made and well fired; sometimes spherical, pyramidal, or discoid loom weights are found in sets, but often the different types may be found together. However, a new type of loom weight – the spool – seems to appear throughout the eastern Mediterranean from the 12th century. Spools can be used for several purposes. They are excellent weights for the warp-weighted loom (Figure 3.9.2), and are highly suited to fine, open weaves, weft-faced weaves, and twills (Andersson Strand and Nosch 2015). Crude loom weights in the shape of spools were identified at 57 sites dating to the last two centuries of the 2nd millennium (Rahmstorf 2011: p. 320). At Lefkandi, spools are dated to Late Helladic (LH) IIIC-early. Lorenz Rahmstorf suggests that the LH IIIC spools at Tiryns may be evidence of changes in the composition of the society. Spools occur sporadically in earlier periods, too, but in LH IIIC they appear in groups, in larger quantities and often with loom weights, in nearly every settlement, but not in graves (Rahmstorf 2003: p. 400). In Tiryns, spools arrive in LH IIIC-middle but most are dated to the LH IIIC-late and to the SubMyc period (Rahmstorf 2003: p. 401). In these layers, no other types of loom weights were unearthed, while in Cyprus, spools appear together with loom weights (Rahmstorf 2003: p. 404; Smith and Tzachili 2012: pp. 145–146). In a few sites in the Aegean (Chania, Sissi, Troy), such spools date back to the 13th century. The spool-shaped loom weights are also common finds in the 11th and 10th centuries in Italy and in central Europe, but are not common in earlier times. Some scholars consider the spools as evidence for Mycenaeans moving toward Cyprus and the Levant, but the place of origin of the objects cannot be determined. Rahmstorf ascribes a common cause to the sudden popularity of eastern Mediterranean clay spools (Rahmstorf 2011: p. 321). It is significant that they first appear toward the end of the Mycenaean Palace period and thus they are not necessarily a Postpalatial phenomenon; in terms of technology, the introduction of spools represents a change of tool, not of technology, since the warp-weighted loom continues in use with these new loom weight types. The appearance of spools may suggest the arrival of new people, with their own tool kits, coming from another region,



Figure 3.9.2 Cypriot plate with depiction of a patterned fabric on a warp-weighted loom with spool-shaped loom weights (Cypro-Geometric [CG] III, c. 850–750 BC; diameter 26 cm). Source: Courtesy of the Akademisches Kunstmuseum Bonn. Reproduced with permission of the Akademisches Kunstmuseum Bonn.

where the warp-weighted loom was also in use. This explanation should be considered in a gendered framework since women appear to be the chief producers of textiles, and the presence of spools would suggest the presence of women among the newcomers. Mounting a loom, warping, and weaving take time, and the spools suggest these women are not simply passing by, but settled.

Textile Terminologies and Clothing Traditions

Mycenaean textile types are quite standardized as illustrated by the textile logograms (Killen 1984; Nosch 2012). Textile and garment terms in Linear B records and in early alphabetical texts and Homer testify both to garment elements that continue from the 2nd to the 1st millennium, and to innovations and changes in clothing. Among the various Mycenaean textiles, some have Greek names, and others are of indeterminate etymology (*tu-na-no*, *pu-ka-ta-ri-ja*, *to-mi-ka*), perhaps of Minoan origin. Mycenaean Greek utilizes the Semitic root *ktn* in the term *ki-to/khiton* (Del Frio, Nosch, and Rougemont 2010). Some Mycenaean textile terms seem to disappear in the 1st millennium, but *ki-to/khitōn*, *we-a₂-no/heanos*, and *pa-wo/pharos* persist. The Mycenaean *pa-wo* and the later Greek *pharos* can be crafted of flax or wool; in the EIA this is a multi-functional fabric for clothing, used as a shroud or for sailcloth, or worn as a purple dyed mantle (*Iliad* 8.221, 24.580, 24.588; *Odyssey* 5.258, 8.84). New clothing terms appear in Greek in the 1st millennium: *peplos*, *chlaina*, *chlamys*, *himation*.

A comparative study of 14th-century clothing traditions in Egypt, the Aegean, central Europe, and Scandinavia indicates similar cloth cultures. Principal elements include wool and linen textiles, mostly woven as plain tabbies and decorated with dyes. The use of fur and skins is seen in footwear and outerwear, mostly sewn from the skins and fur of domesticated animals (Harris 2012). LBA clothing is thus a rich blend of technologies and materials, and nothing suggests that this changed fundamentally in the EIA. In his *Works and Days* (lines 536–546), Hesiod describes male clothing of a Boeotian farmer, c. 700 BC, but a Cretan BA farmer would probably have worn a similar costume:

Then put on, as I bid you, a soft cloak (*chlaina*) and a tunic (*khiton*) to the feet to shield your body – and weave thick wefts on thin warp. In this, clothe yourself so that your hair may keep still and not bristle and stand upon end all over your body. Lace on your feet close-fitting boots of the hide of a slaughtered ox, thickly lined with felt inside. And, when the season of frost comes on, stitch together skins of firstling kids with ox-sinew, to put over your back and to keep off the rain. On your head, wear a shaped cap of felt to keep your ears from getting wet (...) (Translation by H.G. Evelyn-White with minor modifications by the author)

Aspects of continuity and tradition dominate the literature on BA and EIA clothing. Only rarely do scholars identify clothing innovations or the introduction of clothing traditions from outside the Aegean. One of the few

exceptions is the *peplos*, which has been connected to the so-called Dorian migration. The *peplos*, a women's garment type constituting a tubular textile, emerged from a 2nd-millennium central European tradition where it was probably woven on a two-beam loom (Kilian-Dirlmeier 1984a; Hägg 1996). It arrived in Greece at the end of the BA to become an iconic female garment in the EIA and especially in the Archaic (AR) period with its construction as a tubular folded garment fastened on the shoulders.

Bronze dress fasteners, such as fibulae and pins appear in the Aegean from LH IIIB (Desborough 1964: pp. 55–56; Dickinson 2006: pp. 72, 158–164), and their use seems to increase in the EIA, becoming a widespread and integral feature of male and female attire in the Late Geometric (LG) and early AR period (Desborough 1964: p. 57). The increasing use of fibulae is an EIA innovation and reflects a new technique of constructing clothing: whereas BA clothing may have been more fitted to the body by using advanced stitching and tailoring, the use of fibulae suggests an EIA preference for draping cloth around the body. The new emphasis on the use of draped cloth means that clothing becomes a more flexible and multi-functional item that can be draped and used according to social or climatic requirements.

This sartorial turn toward draped clothing fastened by metal devices is also evidenced in Homer. Hera wears a *peplos*, woven by Athena (Pallas Ergane) herself, and fastened with two gold pins, while a *peplos* with two gold pins and six gold fibulae is given to Penelope (*Iliad* 14.178–180; *Odyssey* 18.292–294). Thus, draped clothing with metal dress fasteners had become an elite costume in the EIA.

Fibulae and pins are chiefly known from burials from the Protogeometric (PG) to Middle Geometric (MG) periods. In the LG and AR periods, however, fewer fibulae and pins were deposited in graves, and most are found used as votive offerings especially for female goddesses in certain sanctuaries, e.g. at Lindos and Ialysos on Rhodes, at Pherai in Thessaly, and in the Peloponnese, at the Argive Heraion, in the sanctuaries of Hera Limenia and Artemis Orthia (Kilian 1974: pp. 168–187; Kilian-Dirlmeier 1984b). This suggests that dress fasteners and textiles play a crucial role in EIA Greek cult, either as objects dedicated to the divinities or attached to garments.

It is important to bear in mind that textiles were not only used for clothing, but also for packing, sacks, furnishing, insulation, and bed linen; boats were rigged with sails (*histia*, from *histos* which means loom and also mast) since the MBA. Large quantities of sailcloth were needed for naval fleets, fishing boats, and trade vessels in the Aegean and the Mediterranean of the LBA and the EIA. According to myth, Theseus owned two sets of sails when he sailed between Attica and Crete, a white and a black one, which was an extraordinary extravagance (Barber 1991).

Spinning, weaving, and clothing constitute widely attested and powerful metaphors in Indo-European languages. The Linear B tablets do not permit us to hypothesize whether this was already the case in the LBA, but circumstantial evidence strongly suggests it. Textiles are offered to divinities and sanctuaries in the Linear B tablets, and a female divinity is especially associated with lambs (Nosch 2000). By the EIA, textile patterns are a visually striking element in the decoration of burial vessels, and Homeric Greek is replete with textile expressions: Odysseus is weaving his plot, while the Moirai spin and cut the thread of life. Homeric metaphors use dress to express the emotion of grief, not forgetting the notion of death, as a covering textile or a garment (Cairns 2015). Athena, the goddess of weaving, becomes the major female divinity of the EIA.

Both in the BA Mycenaean textile industries and in EIA Gordion textile workshops, archaeological finds and iconography testify to a much more diverse and sophisticated textile production than we would anticipate from the Linear B texts and logograms and surviving BA textile tools. The pebble mosaic in Gordion Megaron 2 (Figure 3.9.3), dated to the late 9th century, illustrates how the local rulers furnished their palace: the floor must have been covered by many layers of rugs and carpets, all with different woven patterns, in particular the cherished geometric patterns that became so extremely widespread in the EIA.

Comparable ingenious combinations of geometric weaving patterns dominate pottery decoration, especially the lavish funerary vases of EIA Attica and Euboea (Figure 3.9.4). Indeed, the meanders, zigzags, crossed, parallel and straight lines, criss-crosses and triangles all reflect the patterns of tabbies and twills: they break entirely with the rounded and wavy lines and floral and figurative patterns of Minoan and Mycenaean pottery decoration. Weaving patterns seem to be of artistic inspiration for potters/painters for some 200–300 years in the EIA. This hints at an extraordinary creativity in weaving, which generated a wealth of stylized geometric weaving patterns elegantly adapted to round pottery shapes, as if carefully wrapped in a complex woven fabric.

Conclusion

Textile production represents considerable continuity between the BA to the EIA; changes, nevertheless, occur in clothing traditions, such as draped clothing and the use of fibulae, and in weaving tools, such as the spool-shaped loom weight. These changes occur before the beginning of the EIA, and even before the collapse of the Mycenaean palaces. Thus, some typical EIA textile features predate the shift from bronze to iron and introduce new clothing



Figure 3.9.3 Gordion, Megaron 2 (main room): Early Phrygian period pebble mosaic, dated to the late 9th century. Watercolor by Joseph S. Last. Source: Courtesy of the Penn Museum Gordion Archive (Plan 1956–2017, Image 400833). Reproduced with permission of the Penn Museum. (For color version of the figure see color plate section)

traditions as early as the end of LH IIIB. These traditions may be related to changes in the organization and procurement of textile fiber resources at the end of the BA. A case in point is provided by Kastanas, where the LH IIIC-middle witnessed several rapid and related changes, including the increased use of flax, a sudden increase in the height of sheep and their age of slaughter (Becker 1986), and the introduction of spool-shaped loom weights (Rahmstorf 2003: pp. 406–407). The collapse of the Mycenaean palaces would impact on access to fibers and dyes, as well as on cloth production and consumption.

The frequent occurrence of fibulae and pins in graves attests that EIA people were buried in their clothes. Indeed, throughout Greece, the two most



Figure 3.9.4 Attic LG pitcher with water birds on the top rim and geometric patterns similar to weaving. J. Paul Getty Museum. Source: Photograph kindly provided by the J. Paul Getty Museum. Digital image courtesy of the Getty's Open Content Program.

popular grave goods seem to be vessels with woven textile patterns and fibulae and pins as dress fasteners, both textile-related objects. By the beginning of the 1st millennium, new types of draped overclothing fastened with fibulae were being worn, and women began to wear a rather unfitted dress fixed with fibulae. The BA tailored ensembles were abandoned, and EIA clothing became a canvas of geometric weaving patterns displaying aesthetic taste and excellent weaving skills.

REFERENCES

- Andersson Strand, E., and Nosch, M.L. (eds.) (2015). *Tools, Textiles and Contexts: Textile Production in the Aegean and Eastern Mediterranean Bronze Age* (Ancient Textiles Series 21). Oxford: Oxbow Books.
- Apostolakou, V., Brogan, T.M., and Betancourt, P. (2012). The Minoan Settlement on Chryssi and its Murex Dye Industry. In M.-L. Nosch, and R. Laffineur

- (eds.), *Kosmos: Jewellery, Adornment and Textiles in the Aegean Bronze Age, Proceedings of the 13th International Aegean Conference, University of Copenhagen, Danish National Research Foundation's Centre for Textile Research, 21–26 April 2010* (Aegaeum 33). Leuven and Liège: Peeters, pp. 79–182.
- Barber, E.J.W. (1991). *Prehistoric Textiles: The Development of Cloth in the Neolithic and Bronze Ages with Special Reference to the Aegean*. Princeton, NJ: Princeton University Press.
- Becker, C. (1986). *Kastanas: Ausgrabungen in einem Siedlungshügel der Bronze- und Eisenzeit Makedoniens 1975–1979. Die Tierknochenfunde* (Prähistorische Archäologie in Südosteuropa 5). Berlin: Wissenschaftsverlag Volker Spiess.
- Breniquet, C., and Michel, C. (eds.) (2014). *Wool Economy in the Ancient Near East and the Aegean: From the beginnings of Sheep Husbandry to Institutional Textile Industry*. (Ancient Textiles Series 17). Oxford: Oxbow Books.
- Burke, B. (2010). *From Minos to Midas: Ancient Cloth Production in the Aegean and in Anatolia* (Ancient Textiles Series 7). Oxford: Oxbow Books.
- Burke, B. (2007). From The Kingdom of Midas and Royal Cloth Production. In C. Gillis, and M.-L. Nosch (eds.), *Ancient Textiles: Production, Craft and Society. Proceedings of the First International Conference on Ancient Textiles, Held at Lund, Sweden and Copenhagen, Denmark on March 19–23, 2003* (Ancient Textiles Series 1). Oxford: Oxbow Books, pp. 64–70.
- Cairns, D.L. (2015). Clothed in Shamelessness, Shrouded in Grief: The Role of “Garment” Metaphors in Ancient Greek Concepts of Emotion. In G. Fanfani, M. Harlow, and M.-L. Nosch (eds.), *Spinning Fates and the Song of the Loom: The Use of Textiles, Clothing and Cloth Production as Metaphor, Symbol and Narrative Device in Greek and Latin Literature* (Ancient Textiles Series 24). Oxford: Oxbow Books, pp. 25–41.
- Del Frio, M., Nosch, M.-L., and Rougemont, F. (2010). The Terminology of Textiles in the Linear B Tablets, Including Some Considerations on Linear A Logograms and Abbreviations. In C. Michel, and M.-L. Nosch (eds.), *Textile Terminologies in the Ancient Near East and Mediterranean from the Third to the First Millennia BC* (Ancient Textiles Series 8). Oxford: Oxbow Books, pp. 338–373.
- Desborough, V.R. d'A. (1964). *The Last Mycenaeans and Their Successors: An Archaeological Survey, c. 1200 – c. 1000 BC*. Oxford: Clarendon Press.
- Dickinson, O. (2006). *The Aegean from Bronze Age to Iron Age: Continuity and Change Between the Twelfth and Eighth Centuries BC*. Oxford and New York: Routledge.
- Gleba, M. (2017). Tracing Textile Cultures of Italy and Greece in the Early First Millennium BC. *Antiquity* 91 (359), pp. 1205–1222.
- Hägg, I. (1996). Textil und Tracht als Zeugnis von Bevölkerungsverschiebungen. *Archäologische Informationen*, 19, pp. 135–147.
- Harlow, M., Michel, C., and Nosch, M.-L. (eds.) (2014). *Prehistoric, Ancient Near Eastern and Aegean Textiles and Dress: An Interdisciplinary Anthology* (Ancient Textiles Series 18). Oxford: Oxbow Books.
- Harris, S. (2012). From the Parochial to the Universal: Comparing Cloth Cultures in the Bronze Age. *European Journal of Archaeology*, 15, pp. 61–97.

- Kilian, K. (1974). *Fibeln in Thessalien von der mykenischen bis zur archaischen Zeit. Prähistorische Bronzefunde, XIV, 2*. Munich: C.H. Beck.
- Kilian-Dirlmeier, I. (1984a). Der dorische Peplos: ein archäologisches Zeugnis der dorischen Wanderung. *Archaeologisches Korrespondenzblatt*, 14, pp. 281–291.
- Kilian-Dirlmeier, I. (1984b). *Nadeln der frühbelladischen bis archaischen Zeit von der Peloponnes* (Prähistorische Bronzefunde XIII, 8). Munich: C.H. Beck.
- Killen, J.T. (1983). On the Mycenae Ge Tablets. In A. Heubeck, and G. Neumann (eds.), *Res Mycenaeae. Akten des VII. Internationalen Mykenologischen Colloquiums in Nürnberg 1981*. Göttingen: Vandenhoeck and Reprecht, pp. 216–233.
- Killen, J.T. (1984). The Textile Industries at Pylos and Knossos. In C.W. Shelmerdine, and T.G. Palaima (eds.), *Pylos Comes Alive: Industry and Administration in a Mycenaean Palace*. New York: Archaeological Institute of America, pp. 49–63.
- Kolonas, L., Margariti, C., Skals, I., Sarri, K., Vanden Berghe, I., Nosch, M.-L. (2017). Heirs of the Loom: Funerary Textiles from Stamna (Aitolia, Greece). A Preliminary Analysis. In M. Fotiadis, R. Laffineur, Y. Lolos, and A. Vlachopoulos (eds.), *Hesperos. The Aegean Seen from the West. 16th International Aegean Conference, Ioannina, 18–21 May 2016* (Aegaeum 41). Liège: Peeters Publishers, pp. 533–544.
- Kourou, N. (2011). Αττικά τεφροδόχα νεαρών κοριτσιών: Γύρω από έναν αμφορέα και ένα ορυκτοποιημένο ύφασμα της Πρώιμης Γεωμετρικής εποχής. In A. Delivourias, G. Despinis, and A. Zarkadas (eds.), *Έπαινος Luigi Beschi*. Athens: Benaki Museum, pp. 189–200.
- Marinatos, S. (1967). *Kleidung, Haar- und Barttracht. Archaeologia Homerica – Die Denkmäler und das frühgriechische Epos*. Göttingen: Vandenhoeck and Ruprecht.
- Melena, J.L. (1976). La producción de plantas aromáticas en Cnoso. *Estudios Clásicos*, 78, pp. 177–190.
- Michel, C., and Nosch, M.-L. (eds.) (2010). *Textile Terminologies in the Ancient Near East and Mediterranean from the Third to the First Millennia BC* (Ancient Textiles Series 8). Oxford: Oxbow Books.
- Nosch, M.-L. (2000). Schafe unter Potnia und Hermes in Knossos. In F. Blakholmer (ed.), *Österreichische Forschungen zur Ägäischen Bronzezeit 1998* (Wiener Forschungen zur Archäologie 3). Vienna: Phoibos Verlag, pp. 211–215.
- Nosch, M.-L. (2004). Red Coloured Textiles in the Linear B Inscriptions. In K. Staers, and L. Cleland (eds.), *Colour in the Ancient Mediterranean World* (BAR-IS 1267). Oxford: John and Erica Hedges, pp. 32–39.
- Nosch, M.-L. (2012). The Textile Logograms in the Linear B Tablets: Les idéogrammes archéologiques des textiles. In P. Carlier, C. de Lamberterie, M. Egetmeyer, N. Guilleux, F. Rougemont, and J. Zurbach (eds.), *Études mycéniennes 2010. Actes du XIIIe colloque international sur les textes égéens, Sèvres, Paris, Nanterre, 20–23 septembre 2010* (Biblioteca di Pasiphae 10). Pisa: Serra, pp. 305–346.
- Nosch, M.-L. (2015). The Wool Age: Traditions and Innovations in Textile Production and Consumption and Administration in the Late Bronze Age Aegean. In J. Willhartner, and F. Ruppenstein (eds.), *Tradition and Innovation in the Mycenaean Palatial Politics. Proceedings of an International Symposium held at the Austrian*

- Academy of Sciences, Institute for Oriental and European Archaeology, Aegean and Anatolia Department, Vienna, 1–2 March 2013* (Mykenische Studien 34). Vienna: Österreichische Akademie der Wissenschaften, pp. 167–201.
- Nosch, M.-L. (2017). Textile Crops and Textile Labour in Mycenaean Greece. In M. Oller, J. Pàmias, and C. Varias (eds.), *Tierra, territorio y población en la Grècia antiga: aspectos institucionals y míticos*. Mering: Utopica Verlag, pp. 3–38.
- Nosch, M.-L., and Laffineur, R. (eds.) (2012). *Kosmos: Jewellery, Adornment and Textiles in the Aegean Bronze Age, Proceedings of the 13th International Aegean Conference, University of Copenhagen, Danish National Research Foundation's Centre for Textile Research, 21–26 April 2010* (Aegaeum 33). Leuven and Liège: Peeters.
- Rahmstorf, L. (2011). Handmade Pots and Crumbling Loomweights: “Barbarian” Elements in the Eastern Mediterranean in the Last Quarter of the 2nd Millennium BC. In O. Kouka, and V. Karageorghis (eds.), *On Cooking Pots, Drinking Cups, Loomweights and Ethnicity in Bronze Age Cyprus and Neighbouring Regions. An International Archaeological Symposium held in Nicosia, November 6th–7th 2010*. Nikosia: A.G. Leventis Foundation, pp. 315–330.
- Rahmstorf, L. (2003). Clay Spools from Tiryns and Elsewhere. An Indication of Foreign Influence in LH III C? In N. Kyparissi-Apostolika, and M. Papakonstantinou (eds.), *The Periphery of the Mycenaean World. 2nd International Interdisciplinary Colloquium. Lamia 26–30.9.1999*. Athens: Ministry of Culture, ΙΔ' Ephorate of Prehistoric and Classical Antiquities, pp. 397–415.
- Smith, J. (2012). Tapestries in the Mediterranean Late Bronze Age. In M.-L. Nosch, and R. Laffineur (eds.) *Kosmos: Jewellery, Adornment and Textiles in the Aegean Bronze Age, Proceedings of the 13th International Aegean Conference, University of Copenhagen, Danish National Research Foundation's Centre for Textile Research, 21–26 April 2010* (Aegaeum 33). Leuven and Liège: Peeters, pp. 241–250.
- Smith, J., and Tzachili, I. (2012). Cloth in Crete and Cyprus. In G. Cadogan, M. Iacovou, K. Kopaka, and J. Whitley (eds.), *Parallel Lives: Ancient Island Societies in Crete and Cyprus* (BSA Studies 20). Athens: British School at Athens, pp. 141–155.
- Spantidaki, Y., and Moulhéat, C. (2012). Greece. In M. Gleba, and U. Mannering (eds.) *Textile and Textile Production in Europe from Prehistory to AD 400* (Ancient Textiles Series 11). Oxford: Oxbow Books, pp. 182–200.
- Wagner-Hasel, B. (2000). *Der Stoff der Gaben: Kultur und Politik des Schenkens und Tauschens im archaischen Griechenland*. Frankfurt: Campus.